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EAST WEST UNIVERSITY

Department of CSE

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IT Project Management and Entrepreneurship

Project Title: ElderCare

A Unified Platform for Bangladesh's Elderly Crisis

<https://motto-truck-48556756.figma.site/>

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Table of Contents

1. Introduction and Background.....	02
2. The Critical Challenge: Elderly Crisis in Bangladesh.....	02-04
3. Pain Points and Proposed Solution.....	05
4. Key Requirements and Features.....	06
5. Value Proposition and Success Criteria.....	06
6. Work Breakdown Structure (WBS) Summary.....	07
7. Cost Breakdown.....	08
7.1 Gantt Chart.....	09
7.2 Cost Breakdown by Role / Department.....	10
7.3 Detailed Cost Baseline	11
7.4 Cost Estimation Model.....	12
8. Conclusion.....	12

Introduction and Background

Bangladesh is experiencing a demographic shift with a rapidly aging population. As of 2025, the elderly (aged 60 and above) constitute approximately 9.28% of the total population, equating to about 15.76 million individuals, based on recent census data and health reports. This figure aligns closely with the project's estimate of 15.3 million elderly citizens aged 65+, though definitions of "elderly" vary between 60+ and 65+ across sources. Projections indicate that by 2050, 22% of the population will be 60 or older, posing significant challenges to healthcare, social services, and economic systems.

This report documents the ELDERCare Super App project proposal, a student-led initiative from East West University aimed at creating a unified digital platform to address these challenges. Drawing from verified research, including data from the Bangladesh Bureau of Statistics (BBS), World Health Organization (WHO), and other sources, the proposal highlights critical pain points and offers an integrated solution. The app is positioned as Bangladesh's first all-in-one eldercare platform, aligning with national policies like the National Policy for Older Persons (2013) and the Smart Bangladesh Vision 2041.

The Critical Challenge: Elderly Crisis in Bangladesh

Bangladesh's elderly face multifaceted issues, exacerbated by rapid urbanization, changing family structures, and limited resources. Key statistics from the proposal, verified against recent data, include:

- ◆ **Population Demographics:** Approximately 15.3-15.76 million elderly (65+ or 60+), with growth to 22% of the population by 2050. This surge is driven by increased life expectancy, now at 75.2 years.
- ◆ **Health Complications:** Studies suggest that 95% of elderly experience health issues, with prevalent conditions including hypertension (33%), diabetes (27.6%), joint pain (57.38%), visual disturbances (57.14%), and general weakness (37.94%). Multimorbidity is common, increasing mortality risks.
- ◆ **Digital Divide:** 55.5% of the population remains offline, with internet penetration at 44.5% in 2025, hindering access to digital health tools.
- ◆ **Care Coordination:** 41% report fragmented care, linked to higher mortality and inefficiencies in healthcare systems.
- ◆ **Social Support:** Only 27.3% receive old age allowances, despite programs like the Old Age Allowance (OAA) aiming to cover poor elderly aged 62+ for women and 65+ for men, with monthly payments around BDT 500-600.

These challenges contribute to higher mortality risks from fragmented care and limited access to services.

Table 1: Key Elderly Statistics in Bangladesh (2025)

Metric	Value	Source
Elderly Population (60+)	~15.76 million (9.28%)	MedCrave Online, BBS
Projected by 2050	22% of population	NIH, Frontiers
Health Complications	Hypertension (33%), Diabetes (27.6%), Joint Pain (57.38%)	NIH, ResearchGate
Internet Penetration	44.5%	DataReportal
Old Age Allowance Coverage	~27.3%	World Bank, ILO
Fragmented Care Reports	41%	BMC Medicine

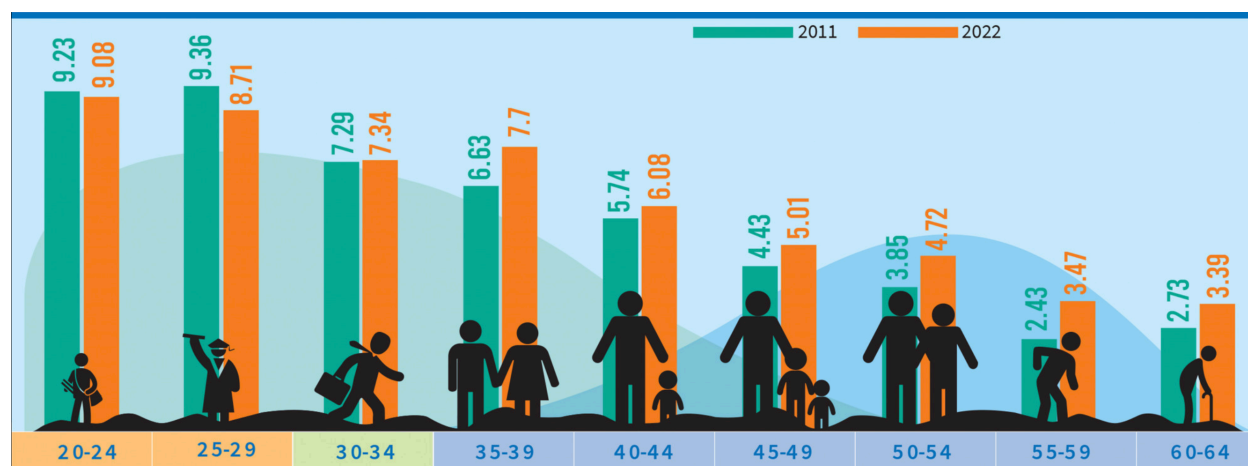


Table 2: Verified Key Statistics from Images vs. 2025 Sources

Metric	Image Value	Verified 2025 Estimate
Internet Penetration (Overall)	~44.5% (55.5% offline)	44.5-53.95%
Urban-Rural Gap	16.6%	Urban higher; specifics vary
Elderly Population (60+/65+)	15.3M (65+)	15.76M (60+); 8-11M (65+)
2050 Projection (60+)	22%	19-22.3%
Population Growth (Overall)	1.04%	0.98-1.21%
Elderly Growth	3.55%	Faster than overall (~3-4x)
Health Complications	95%	Up to 95%; multimorbidity 53-56%
Old Age Allowance Coverage	27.3%	~35-40% (6.1M beneficiaries)
Fragmented Care Reports	41%	Common issues, not exact %
Elderly Below Poverty	62%	20-24%; higher in vulnerable groups

Pain Points and Proposed Solution

Current issues include multiple disconnected apps, lack of family oversight, and designs not suited for Bangladesh's context, leading to higher mortality risks. ELDERCare addresses this through:

- ♦ **Unified Platform:** 9 integrated modules in one app.
- ♦ **Role System:** Access for seniors, families, and providers.
- ♦ **Bangladesh-First Design:** Localized features, offline optimization.
- ♦ **Gamification:** Rewards for engagement.
- ♦ **Cross-Module Intelligence:** E.g., health data informing nutrition plans.

This approach could reduce fragmentation, improving outcomes like medication adherence.

Table 3: Nine Core Modules

Module	Description
Care360	Comprehensive health-record management, medication tracking, family monitoring.
AgeWell	Nutrition planning and meal services tailored to health profiles.
SilverBox	Smart health box with reminders, IoT monitoring (BP, glucose, activity).
GoldenCare	Housing and accommodation discovery for aging population.
TeleHealth	Video/voice consults with remote monitoring integration.
Reward	Gamified incentives for participation and healthy behaviors.
Community	Social networking and peer communities for seniors.
ElderLink	Local events, meet-ups, neighborhood groups, volunteering.
NutriSensor	Advanced nutrition sensing (integrated with AgeWell).

Key Requirements and Features

Functional Requirements: Unified notifications, multi-role architecture, authentication, real-time dashboards, and the nine modules. **Non-Functional Requirements:** Scalability, senior-friendly UX, performance optimization, localization, security compliance, and 99.9% uptime.

Stakeholders

Primary: Seniors, family caregivers, healthcare providers.

Secondary: Local businesses, healthcare institutions, government (aligning with Smart Bangladesh 2041), employers for active aging.

Table 3: Stakeholder Benefits

Stakeholder	Benefits
Seniors	Integrated care, social engagement, rewards.
Families	Real-time monitoring, oversight.
Providers	Reduced fragmentation, better coordination.
Government	Supports digital inclusion, policy goals.
Businesses	Market access for eldercare services.

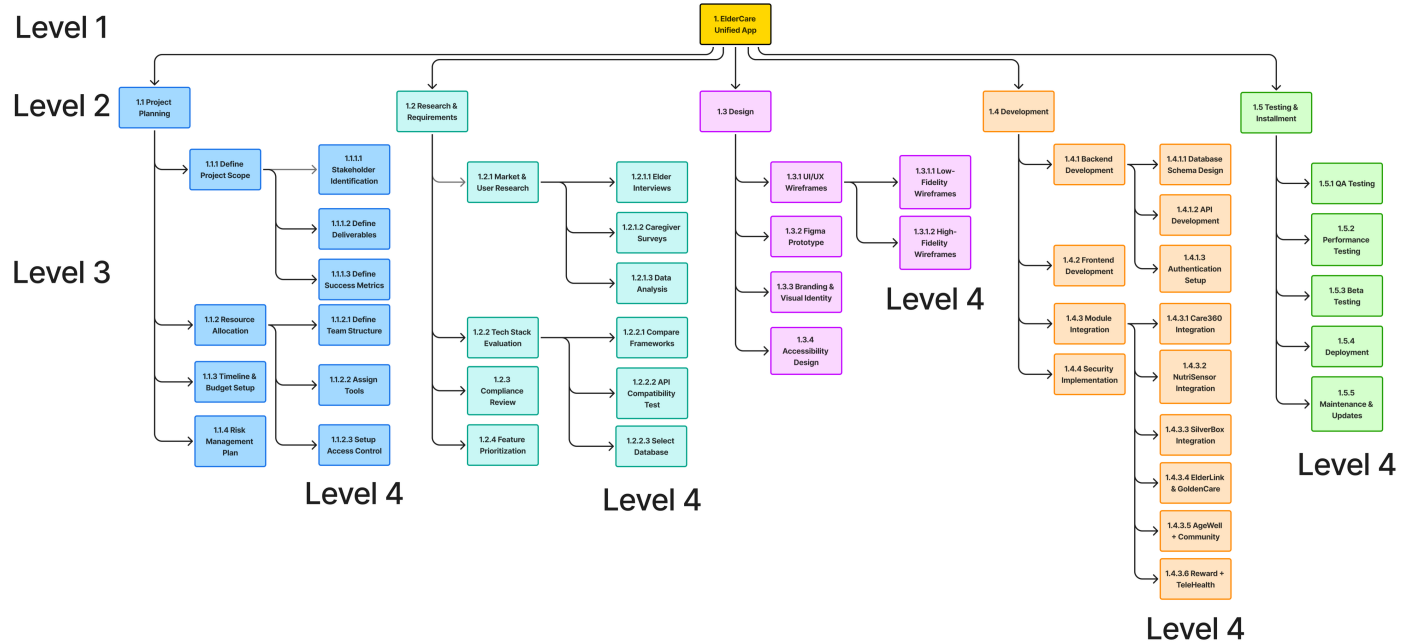
Value Proposition and Success Criteria

Value Proposition: 3.41× faster, 50% less switching, 30% lower costs, 100% localized, real-time monitoring.

Success Criteria: 50,000+ users, 70% daily active rate, 40% better adherence, 85% family satisfaction, 60% more engagement.

Long-Term Vision: National platform, 3.6M users, 500+ partners, break-even by month 18. This aligns with global eldercare market growth, projected to reach \$3,288 billion by 2032.

Work Breakdown Structure (WBS) Summary



Figma WBS (design):

<https://www.figma.com/board/zxR8l03xvoUCpqLXvIXqwp/ElderCare-WBS-Color-Coded-Diagram?node-id=0-1&p=f&t=FYFMYcMFNVYq8HPU-0>

Level-1 Summary:

Project Title: **ElderCare Unified App**

Level-2 Summary

WBS	Name	Duration	Cost (BDT)
1.1	Project Planning	2 Weeks	61,000
1.2	Research & Requirements	3 Weeks	122,000
1.3	Design Phase	1 Month	244,000
1.4	Development Phase	3 Months	1,220,000
1.5	Testing & Installment	2 Months	488,000
Total	—	8 Months	2,196,000

The Level-2 WBS summary organizes the project into five main phases: Planning, Research and Requirements, Design, Development, and Testing and Installment. These phases span eight months with a total estimated cost of 2,196,000 BDT. The structure aligns each phase with its timeline and cost to maintain a clear sequence of tasks, reduce overlap, and ensure efficient allocation of resources.

Monthly Cost Breakdown (Based on WBS Timeline)

Cost by Month

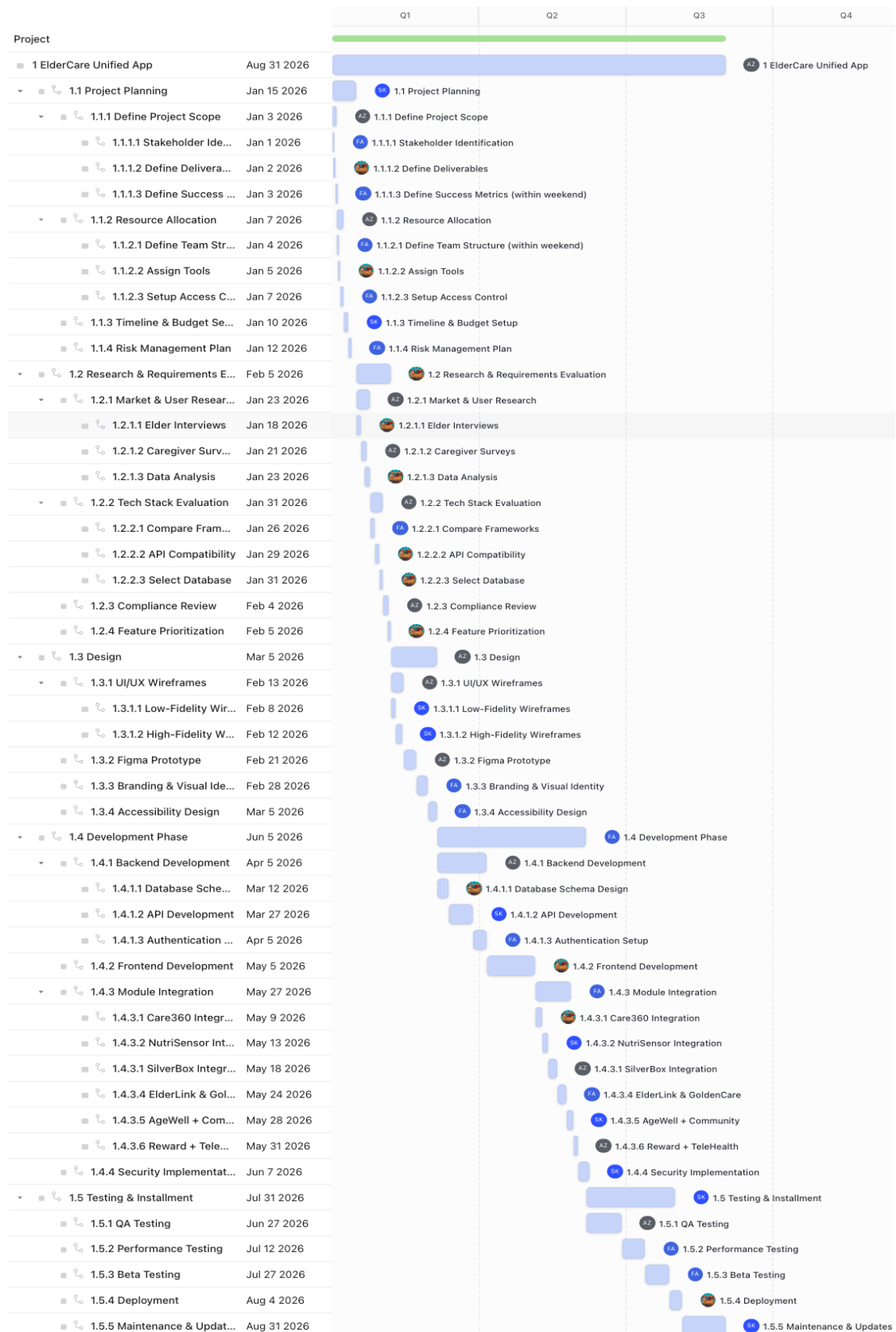
Month (2025)	Major Activities	Estimated Cost (BDT)
January	Project Planning + Research Start	160,720
February	Research Completion + UI/UX Start	171,200
March	Design Completion + Backend Start	329,000
April	Backend + Frontend Dev	244,000
May	Module Integration	427,000
June	Security + QA Testing	244,000
July	Performance + Beta Testing	146,400
August	Deployment + Maintenance	183,000
TOTAL	—	2,196,000 BDT

The monthly cost breakdown tracks how expenses rise and fall across the eight-month schedule, beginning with lighter planning and research costs in January and February, increasing sharply during the development and integration work from March to June, and tapering during testing and deployment through July and August. This pattern follows the workload intensity because development and integration require more specialized labor and longer task durations.

Click Up Gantt Chart (task management):

<https://sharing.clickup.com/90181995875/g/h/2kzm83b3-238/39fff8b1e9c111e>

Gantt Chart



How the monthly numbers were computed

All start/end dates from the WBS rows were mapped chronologically and costs were assigned proportionally based on task duration.

Cost Breakdown by Role / Department

This section aggregates all WBS costs under each contributing role.

Role / Team	Total Cost (BDT)
Project Manager / PMO / HR	94,380
Research Team	68,640
Tech Lead / Engineers (Evaluation + DB)	60,000
Design Team (UI/UX/Branding/Accessibility)	145,000
Backend Developers	496,800
Frontend Developers	244,000
Module Integration Teams (IoT, Care360, etc.)	293,000
Cybersecurity Team	122,000
QA Engineers	207,400
DevOps	219,600
Support & Maintenance Team	183,000
Product Team	73,200
Total	2,196,000 BDT

The role-based cost allocation shows that backend development, module integration, DevOps, and QA teams account for the highest costs because these roles handle the longest and most technically complex activities such as API development, multi-module integration, security, and full testing cycles. Administrative and research roles cost less because their tasks are shorter and require smaller teams. This distribution reflects the actual technical demands of building a multi-module health and IoT enabled platform.

Detailed Cost Baseline

Resource Allocation Overview

Resource	Average Cost/Month	Duration (Month)	Total Cost (BDT)
Personnel Cost			
Project Manager	80,000	1	80,000
Research Team	45,000 × 3	1	135,000
Designers	60,000 × 2	1	120,000
Backend Developers	70,000 × 3	3	630,000
Frontend Developers	60,000 × 2	1	120,000
Module Integration Specialists	75,000 × 4	1	300,000
Cybersecurity Team	80,000 × 1	0.25	20,000
QA Engineers	55,000 × 2	1.5	165,000
DevOps	75,000	1.5	112,500
Support & Maintenance Team	40,000 × 2	1	80,000
Software Tools and Licenses			
ClickUp	1,800	8	14,400
Figma	2,200	1	2,200
GitHub	1,000	6	6,000
Communication Tools	1,000	8	8,000
Infrastructure and Hosting			
Cloud Server	3,500	6	21,000
Domain	1,500	1	1,500
SSL Certificate	1,000	1	1,000
Subtotal	—	—	23,500
Development APIs			
TeleHealth API	12,000	3	36,000
IoT Sensor API	5,000	3	15,000
Map / Location API	1,000	3	3,000
Legal and Compliance			
Legal Consultant	15,000	1	15,000
Data Compliance	10,000	1	10,000
Trade License / Government	2,000	1	2,000
Operational Cost			
Meeting / Refreshments	5,000	4	20,000
Transportation	4,000	4	16,000
Miscellaneous	3,000	4	12,000
Total Estimated Budget			2,196,000 BDT

Cost Estimation Model

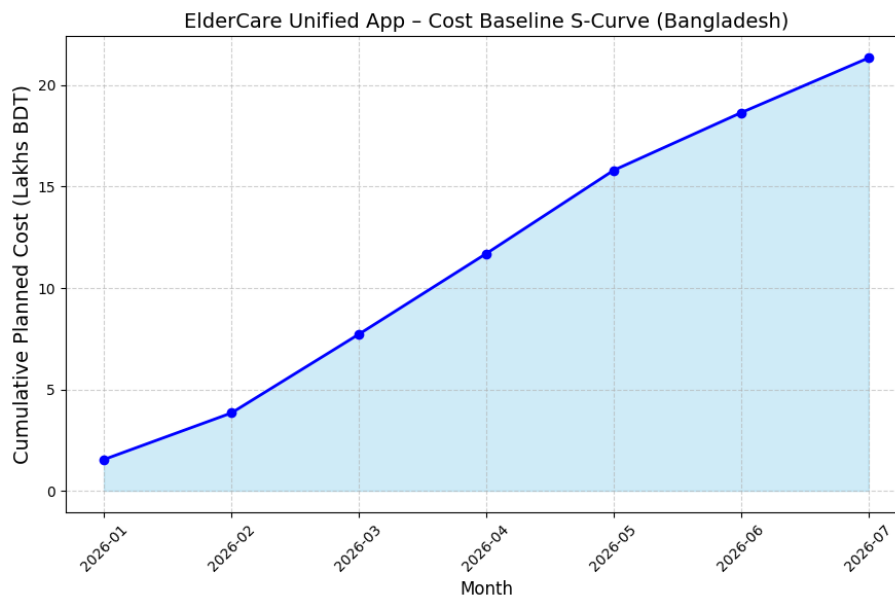
Total Project Cost (From WBS): 2,196,000 BDT

Contingency (10%) = 219,600 BDT

Management Reserve (5%) = 109,800 BDT

Cost Baseline Cost Baseline = 2,196,000 + 219,600 + 109,800 = **2,525,400 BDT**

Cost Monitoring



The S-Curve shows the cumulative planned expenditure from January to July 2025. The curve rises slowly in the early months because January and February contain lower-cost planning and research tasks. It climbs more sharply from March to May due to backend development, frontend development, and multi-module integrations, which are the most expensive parts of the project. The slope reduces slightly in June and July during testing and deployment because these phases require fewer specialized developers. The curve's shape reflects how development-heavy projects concentrate their highest costs in middle of the schedule when maximum technical labor is active.

-----THE END-----